VLAN Concepts and Configuration

Cisco Packet Tracer Lab Report

1. Theory: VLANs

A VLAN (Virtual Local Area Network) logically divides one physical switch into several separate broadcast domains. Devices in different VLANs cannot talk to each other at Layer 2; traffic between them must pass through a router or Layer 3 switch.

- Segmentation: groups devices by function (department, lab) instead of physical location.
- Benefits: smaller broadcast domains, better security, easier management, and lower cost than buying separate switches.
- Use cases: separate departments, labs, guest Wi-Fi, voice (IP phones), and server networks.

2. Task 1: Create the VLANs

Create VLAN 10 named Engineering and VLAN 20 named Marketing on the switch:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Engineering
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Marketing
Switch(config-vlan)# exit
```

3. Task 2: Assign the Ports

FastEthernet0/1 and 0/2 go to VLAN 10; FastEthernet0/3 and 0/4 go to VLAN 20:

```
Switch(config)# interface range fastEthernet 0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
Switch(config)# interface range fastEthernet 0/3 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# end
Switch# copy running-config startup-config
```

Verify with: show vlan brief

```
VLAN Name         Status    Ports
---- ------------ --------- ---------------------
1    default      active    Fa0/5 ... Fa0/24, Gi0/1, Gi0/2
10   Engineering  active    Fa0/1, Fa0/2
20   Marketing    active    Fa0/3, Fa0/4
```

4. Task 3: VLAN Configuration Record

VLAN ID	VLAN Name	Assigned Ports	Port Mode
10	Engineering	FastEthernet0/1, FastEthernet0/2	Access
20	Marketing	FastEthernet0/3, FastEthernet0/4	Access
1	default	All other ports (Fa0/5 onward)	Access

Test: PCs in the same VLAN can ping each other (for example, two PCs on Fa0/1 and Fa0/2 in the same subnet). A PC in VLAN 10 cannot ping a PC in VLAN 20 without inter-VLAN routing.

5. Task 4: Benefits of VLANs in a College Campus

Consider a campus with separate labs for coding and networking:

1. **Better security and isolation.** Placing the coding lab and the networking lab in different VLANs keeps their traffic apart. Students experimenting with routers, scanning tools or misconfigured devices in the networking lab cannot disturb or snoop on the coding lab's machines and servers, and access between the labs can be controlled in one place.
2. **Reduced broadcast traffic and better performance.** Each VLAN is its own broadcast domain, so broadcasts from one lab (such as DHCP, ARP or routing protocol traffic from lab experiments) do not flood the other lab. This keeps the network faster and more stable. VLANs also make it easy to move or regroup labs through configuration instead of re-cabling.

6. Conclusion

VLANs 10 (Engineering) and 20 (Marketing) were created and assigned to their ports on the switch, and the configuration was verified with show vlan brief.